CRMPro AI

Developer Technical Specification

Full product implementation blueprint • Business modules • UI pages • Data model • APIs • AI/RAG • Security • Testing • Micro-phases

Selected baseline

Angular 20+ | ASP.NET Core 10 Web API | PostgreSQL + pgvector | EF Core Code First | Clean Architecture | Modular Monolith

Prepared for a complete, portfolio-grade B2B CRM for IT services and software development companies. Designed for incremental implementation using AI-assisted / vibe coding with human review, tests and controlled commits.

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Source baseline: CRMPro AI Business Requirements Document (BRD), Version 1.0, 34 pages.

Document control and how to use this specification

Item	Definition
Product	CRMPro AI – multi-tenant B2B CRM for IT services/software development firms.
Audience	Product owner, developer, AI-assisted coding agent, QA/tester, DevOps engineer and technical reviewer.
Status	Developer baseline. Business decisions marked OPEN must be confirmed before production.
Implementation style	Small vertical slices. Each phase must compile, pass tests, include migrations where needed and be demoable.
Architecture	Clean Architecture + DDD-informed modular monolith + CQRS-style application use cases.
Database	PostgreSQL; EF Core Code First migrations. pgvector enabled for semantic retrieval.
Frontend	Angular 20+, standalone components, PrimeNG, Tailwind CSS, Signals/RxJS.
Export	Excel (.xlsx) only for user-facing tabular exports. No CSV/PDF export unless explicitly approved later.

How a developer or coding agent must use this document

- Treat BRD requirements as the business baseline. This technical specification adds proposed implementation details where the BRD intentionally leaves UI fields, API contracts and schema choices open.
- Do not silently change a business rule, status, permission, workflow or field definition. Raise an OPEN decision and record the approved answer.
- Implement one micro-phase at a time. Do not ask an AI coding agent to build an entire module in one prompt.
- Before editing, inspect the existing solution and current phase state. Preserve conventions already established unless a documented architecture decision changes them.
- Every phase must include code, tests, migration (if schema changes), authorization, tenant isolation, audit behavior, UI loading/error/empty states and updated documentation.
- Never accept generated code without compiling, reviewing the diff, running tests and manually exercising the workflow.

Important distinction: The BRD defines business capabilities and priorities. Exact grid columns, search filters, form layouts, endpoint shapes and entity property names below are implementation proposals to make the product buildable. Confirm configurable values with the product owner.

Contents

- 1. Product purpose, boundaries, stakeholders and business lifecycle
- 2. Stack, architecture, solution structure and engineering standards
- 3. Tenancy, identity, authorization, audit and common UI conventions
- 4. Module specifications: pages, grids, filters, forms, actions, workflows and acceptance
- 5. Customer portal and reporting
- 6. AI platform: model gateway, prompts, tools, RAG, agents, evaluations and governance
- 7. Data model, API conventions, integrations, background processing and operations
- 8. Testing, security, deployment, demo readiness and acceptance
- 9. Very small implementation phases and completion gates
- 10. Open business decisions and developer checklists

1. Product definition and business purpose

CRMPro AI is a multi-organization SaaS CRM for small and medium-sized IT services and software development companies. It manages the relationship from first enquiry through qualification, opportunity, proposal, customer onboarding, account management and post-sale support. The product centralizes records and interaction history and adds permission-aware AI assistance.

Business outcomes

Outcome	Product capability	Measures / evidence
Better lead response	Capture enquiries, assign owners, make follow-ups visible.	Time to first meaningful response; active leads with owner and next action.
Improved sales execution	Stages, deal values, risks, next steps and pipeline visibility.	Win rate; stage ageing; sales cycle; close-date hygiene.
Consistent customer knowledge	Connected accounts, contacts, proposals, activities, tickets and documents.	Completeness of key account data; visible customer timeline.
More effective support	Ticket queues, priority, SLA, escalation and reusable knowledge.	First response; resolution time; SLA attainment; reopen rate.
Less repetitive work	AI summaries, drafts, retrieval and bounded workflows.	Draft review/acceptance; time saved; tool workflow success.
Management visibility	Consistent dashboards and evidence-based reports.	Reconciled KPIs; forecast quality; report adoption.

Target organization and users

Planning assumption: an IT services company with approximately 5–200 internal users, B2B sales, repeat customer relationships and a support/maintenance function. This is a design assumption, not a hard technical limit.

Stakeholder / role	Primary goals	Typical responsibilities
Business owner / executive	Revenue, customer health and operations visibility.	Review dashboards; approve key commercial decisions; monitor performance.
Organization administrator	Configure CRM and control access.	Users, roles, teams, settings, categories, AI/data policies.
Sales manager	Improve pipeline and conversion.	Assign leads; review deals; coach sales team; approve proposals where required.
Sales executive	Convert enquiries into business.	Qualify leads; manage contacts/opportunities; schedule follow-ups; prepare proposals.
Account manager	Maintain and grow customers.	Customer reviews; relationship risks; expansion opportunities.
Support manager	Effective support operation.	Queues, assignment, escalation, SLA policy and support performance.
Support agent	Resolve customer issues.	Work assigned tickets; use knowledge; record diagnosis and resolution.
Customer portal user	Submit and monitor support requests.	Create tickets; add replies/attachments; view permitted tickets and shared knowledge.
Platform operator	Operate SaaS platform.	Availability, subscription/service status and platform support without default access to tenant business content.

Core domain vocabulary

Object	Meaning	Important distinction
Lead	Unqualified or not-yet-converted interest from a person/company.	Not automatically a customer or deal.
Account	Organization with a prospect/customer relationship.	Can exist before a sale; many contacts and opportunities.

Object	Meaning	Important distinction
Contact	Person associated with an account or lead.	May be decision-maker, technical evaluator, billing or primary contact.
Opportunity / Deal	Specific potential commercial engagement.	One account may have multiple opportunities.
Proposal	Versioned commercial response to an opportunity.	Approval and sending history must be retained.
Activity	Task or interaction: call, email, meeting, demo or follow-up.	Link to relevant lead/account/contact/opportunity/ticket.
Ticket	Customer issue, question, request or incident.	Owner, priority, status, communications and resolution.
Knowledge article	Approved reusable answer, instruction, FAQ or policy.	Only published and authorized content is eligible for RAG.
Tenant	Independent company using CRMPro AI.	No tenant can access another tenant's records.

Product boundary

Included	Excluded from initial scope
Tenant/user administration; leads; accounts/contacts; opportunities; activities; proposals; tickets; customer portal; knowledge/document management; dashboards/reports; AI assistant and bounded workflows; audit; notifications; import/export; basic integrations.	Accounting/ledger/tax/payroll/invoicing; full project delivery/sprints/source control/time tracking; full marketing automation; telephony/call recording; social publishing; e-signature; autonomous negotiation/pricing/contracts; full ERP/procurement/inventory/manufacturing/field service.

End-to-end relationship lifecycle

- Awareness/enquiry → lead captured with source and owner.
- Qualification → qualify, disqualify or nurture; record fit, need, budget/authority/timing and next action.
- Opportunity → create deal linked to account and primary contact; stage, value, currency, close date and owner.
- Proposal/negotiation → create versioned proposal, review/approve, send and record response.
- Won/customer activation → record outcome date/final value, update account relationship to Customer and initiate handoff.
- Account management → timeline, customer reviews, service interests, risks and expansion opportunities.
- Support/maintenance → ticket created, assigned, responded to, resolved and retained in account history.
- Renewal/expansion → create a new opportunity; retain the account relationship.

2. Technology stack and architecture

Concern	Selected technology / approach	Implementation notes
Frontend	Angular 20+	Standalone components, lazy routes, Reactive Forms, Signals + RxJS, HttpClient, route guards.
UI	PrimeNG + Tailwind CSS	PrimeNG tables, dialogs, menus, forms, date pickers, charts; shared design tokens.
Backend	ASP.NET Core 10 Web API	REST APIs, OpenAPI/Swagger, ProblemDetails, validation, rate limiting.
Architecture	Clean Architecture + modular monolith	Domain, Application, Infrastructure, API; module boundaries enforced by architecture tests.
Use-case pattern	CQRS-style commands/queries	MediatR optional; use explicit handlers and interfaces; avoid ceremony for trivial code.
Persistence	PostgreSQL + EF Core 10	Code First migrations; explicit indexes/constraints; concurrency tokens.
Vector search	pgvector	Embedding storage and similarity search; metadata and permission filters.
Authentication	ASP.NET Core Identity + JWT/refresh tokens or OIDC	Choose final identity deployment model before implementation. Never build custom password crypto.
Cache	Redis	Cache keys must include tenant and authorization scope; invalidation is explicit.
Async messaging	RabbitMQ + Outbox	Document ingestion, notifications and durable background workflows.
Background work	Dedicated .NET Worker Service	Idempotent consumers; retry/dead-letter handling; job status persisted.
AI abstractions	Microsoft.Extensions.AI	Provider-neutral IChatClient / embedding abstractions; adapter per provider.
AI orchestration	Semantic Kernel selectively	Use when planner/plugin/process capabilities justify it; do not make it mandatory for simple calls.
Files	S3-compatible object storage	Store object keys and metadata in DB; never trust client filename/path.
Observability	Serilog + OpenTelemetry	Structured logs, traces, metrics; redact sensitive prompts and content.
Testing	xUnit, FluentAssertions, Testcontainers, Playwright	Add Angular unit/component tests with the chosen Angular test runner.
Delivery	Docker Compose, GitHub Actions, AWS	Local reproducible stack; CI gates; secrets in managed secret store.

Architectural style

Use a modular monolith for the first complete product. It gives clear module boundaries without distributed-system overhead. Keep modules independently testable and avoid cross-module table access from application code. Extract a service only when a demonstrated scaling, ownership or deployment need exists.

Dependency rules

- Domain: entities, value objects, aggregates, domain events, domain policies. No EF Core, ASP.NET, Angular or provider SDK references.
- Application: use cases, commands/queries, DTOs, validators, authorization requirements and interfaces/ports. Depends on Domain.
- Infrastructure: EF Core mappings, DbContext, repositories, storage, email, Redis, RabbitMQ, AI provider adapters. Implements Application ports.
- API: HTTP endpoints/controllers, authentication middleware, request/response mapping, DI composition root. Calls Application; must not contain business rules.

- Worker: consumes durable messages and invokes Application use cases; must establish tenant and actor context explicitly.
- Frontend: feature areas call typed API clients; no direct database access; do not duplicate business rules as the source of truth.

Suggested solution structure

```
src/
CRMProAI.Api/
CRMProAI.Application/
Common/ (Behaviors, Abstractions, Validation, Authorization)
Modules/ (Identity, Tenancy, Leads, Accounts, Sales, Activities, Proposals, Support, Knowledge, Reporting, AI,
Audit)
CRMProAI.Domain/
Common/ (Entity, AggregateRoot, ValueObject, DomainEvent)
Modules/ (module-owned domain model)
CRMProAI.Infrastructure/
Persistence/ (DbContext, Configurations, Migrations)
Identity/ Storage/ Messaging/ AI/ Email/ Caching/
CRMProAI.Worker/
CRMProAI.Contracts/ (only stable cross-boundary contracts)
tests/
CRMProAI.Domain.UnitTests/
CRMProAI.Application.UnitTests/
CRMProAI.Infrastructure.IntegrationTests/
CRMProAI.Api.IntegrationTests/
CRMProAI.ArchitectureTests/
web/crmpro-ai-web/
src/app/core/ shared/ features/ layouts/ api/
web/crmpro-ai-portal/ (separate app; may be introduced with support release)
deploy/ docker/ scripts/ docs/
```

Module boundaries

Module	Owns	Publishes / consumes
Identity & Access	Users, roles, permissions, teams, invitations, refresh sessions.	User/role changes; authorization checks.
Tenant & Organization	Tenant profile, settings, business hours, branding, feature flags.	Tenant provisioned/configuration changed.
Lead Management	Lead, source, qualification, conversion, duplicate review.	Lead qualified/converted/assigned.
Accounts & Contacts	Account, contact, relationship, timeline links.	Account/contact created or updated.
Sales & Opportunities	Pipeline, stages, transitions, loss reasons, outcomes.	Opportunity stage/won/lost.
Activities	Tasks, meetings, calls, reminders, interaction log.	Activity due/completed.
Proposals	Templates, proposal, versions, approval and send history.	Proposal approved/sent/accepted.
Support	Ticket, comments, attachments, queues, SLA, resolution.	Ticket assigned/escalated/resolved.
Knowledge	Articles, documents, versions, access classification, ingestion state.	Content published/withdrawn; re-index requests.
Reporting	Approved KPI definitions, report queries and exports.	Read models / reporting refresh.
AI	Conversations, prompts, model calls, embeddings, retrieval, tool runs, approvals, evaluations, usage.	AI run status; approved action execution.
Audit & Notifications	Audit events, notification preferences, delivery attempts.	Domain events to audit/notify.

Cross-module communication: use Application contracts and domain/integration events. Use an Outbox table in the same transaction for events that must be published reliably. Do not let one module write another module's tables.

3. Common engineering standards

Multi-tenancy and authorization

- Every tenant-owned aggregate/table must have TenantId (UUID) and tenant-scoped unique constraints/indexes. Platform-level records are explicitly marked as global.
- Resolve tenant from a trusted authenticated claim or validated host/subdomain mapping; never trust a client-supplied TenantId as authorization.
- Apply tenant query filters as a convenience, not the sole security boundary. Add PostgreSQL Row-Level Security as defense in depth where feasible; set tenant context safely per DB transaction/connection.
- All background jobs, message consumers, exports, search and AI retrieval must carry tenant context and re-check permissions.
- Record-level policies: owner, team, assigned queue, account visibility and role permissions. Deny by default.
- Customer portal authorization is organization-scoped and separately checks contact/ticket visibility. Never expose internal notes, private attachments or commercial data.
- Platform operator must not browse tenant business content by default. Any break-glass support access must be time-bound, approved, reason-coded and audited.

Identity and permission model

Role template	Default access profile
Organization Admin	Manage tenant configuration, users, roles, permissions and business settings; business data access configurable.
Sales Manager	Team/organization sales visibility; assign leads; manage/review opportunities; approve proposals as permissioned.
Sales Executive	Own/permitted leads and opportunities; create/edit sales records; no broad support administration.
Account Manager	Assigned accounts and related relationship history; create reviews and expansion opportunities.
Support Manager	Support queues, SLA policy, assignments, escalation, support reports and knowledge management.
Support Agent	Assigned/queue tickets and required customer context; no unrelated commercial pipeline by default.
Knowledge Administrator	Create, review, publish, version and classify knowledge content.
Customer Portal User	Own organization’s permitted tickets, replies, attachments and shared/published knowledge only.
Platform Operator	Platform health and subscription operations; no tenant content by default.

Implement permission keys such as leads.read, leads.create, leads.update, leads.delete, leads.export, leads.assign, opportunities.stage.change, proposals.approve, tickets.internal-note.read, tickets.reply.public, knowledge.publish, ai.use, ai.action.approve, audit.read and users.manage. Combine role grants with resource-level policies.

Common list-page conventions

Element	Required behavior
Page header	Breadcrumb, page title, concise description, primary Add New button where permitted.
Toolbar	Search box, filters, column chooser (optional), refresh, Export Excel. Export permission checked server-side.
Grid	Server-side pagination, sorting, filtering; stable row key; loading skeleton; empty state; error state; accessible column labels.
Row actions	View, Edit, Delete/Archive, Assign, Convert or other module-specific actions. Show only allowed actions; API rechecks permission.
Delete behavior	Prefer archive/soft delete for business records. Hard delete restricted to retention/privacy workflows. Confirm impact and dependent links.

Element	Required behavior
Bulk actions	Only when explicitly supported; show selected count and preview; enforce permissions and audit.
Excel export	.xlsx only; export current filtered dataset or explicitly selected rows; tenant and row permissions enforced; audit event; protect against formula injection.
Forms	Reactive Forms; required markers; field-level errors; server validation surfaced; prevent double submit; unsaved-change warning.
Detail page	Summary header, status/owner, action menu, tabs for Overview, Activity/Timeline, Related records, Files, Audit where permitted.
Responsive/accessibility	Keyboard accessible; visible focus; sufficient contrast; responsive layout; meaningful labels and validation.

Common record lifecycle and audit

- Use CreatedAtUtc, CreatedByUserId, UpdatedAtUtc, UpdatedByUserId, TenantId, RowVersion/concurrency token and IsDeleted/ArchivedAtUtc where applicable.
- Audit important changes with entity type/id, action, actor, tenant, timestamp, correlation id, changed-field summary and reason when required. Do not store secrets or full sensitive prompts in audit payloads.
- Status transitions must be validated by application/domain rules, not just disabled UI controls.
- Use UTC in storage and APIs; display in tenant/user timezone. Currency is explicit per opportunity/proposal; do not sum unlike currencies without conversion rules.
- Use soft-delete/archive for leads, accounts, contacts, opportunities and articles; block deletion where dependent business history exists unless a reviewed merge/retention workflow is used.

4. Tenant, organization and identity modules

Business purpose: Provision an isolated company workspace and administer its users, access model and CRM configuration.

Primary stakeholders: Platform operator, organization administrator, business owner and all internal users.

Expected outcome: Each company can securely configure its workspace, invite users and operate with role-appropriate access.

AI involvement: AI model/provider configuration, feature access and usage budgets are tenant-governed. Tenant configuration must scope all AI calls and logs.

4.1 Tenant provisioning / organization profile

Purpose: Create or provision the tenant and maintain organization details.

List/grid columns: Tenant name, tenant key/slug, legal name, website, industry, primary admin, plan/status, created date.

Filters/search: Tenant status, tenant name/slug, created date, plan (if subscriptions enabled).

Toolbar and row actions: Platform-only: Provision tenant, Suspend/Reactivate, View operational status. No tenant business-data browsing.

Create/edit form: Organization name, legal name, slug (unique), website, industry, address, default timezone, default currency, locale, logo, support email, primary admin.

Detail view/tabs: Tabs: Profile, Branding, Business Settings, Integrations, AI Governance, Usage. Separate platform admin screen from tenant admin screen.

Validation/business rules: Slug unique globally; tenant status Active/Suspended/Provisioning; platform operator actions audited.

4.2 Users

Purpose: Invite and manage employees and portal identities.

List/grid columns: Name, email, username, role(s), team, status, last login, invited date.

Filters/search: Name/email search, role, team, status, last login range.

Toolbar and row actions: Invite User, Resend Invite, Edit, Activate/Deactivate, Reset/Send Password Reset, Assign Roles, View Audit. Export Excel permission-controlled.

Create/edit form: First/last name, email, phone optional, role templates, teams, manager, timezone, locale, status, portal/internal user type. Invitation expires; user sets password through secure flow.

Detail view/tabs: User detail: profile, roles/permissions, teams, active sessions, audit. Portal users linked to account/contact.

Validation/business rules: Unique normalized email per tenant unless product identity design chooses global identity; deactivation revokes sessions; cannot remove last active tenant admin.

4.3 Roles and permissions

Purpose: Configure permission bundles without hard-coding roles into UI.

List/grid columns: Role name, description, system/custom, user count, permission summary, modified date.

Filters/search: Role name, system/custom, permission category.

Toolbar and row actions: Create Role, Clone Role, Edit, View Users, Deactivate. System roles cannot be deleted.

Create/edit form: Role name, description; permission matrix grouped by module/action; optional scope (own/team/organization/assigned queue).

Detail view/tabs: Detail: permission matrix and assigned users.

Validation/business rules: Least privilege defaults; prevent privilege escalation; permission changes audited; existing sessions/claims refreshed.

4.4 Teams and business settings

Purpose: Configure sales/support teams, pipeline values, categories, priorities, SLA calendars and defaults.

List/grid columns: Team name, team type, manager, members count, active status.

Filters/search: Team name/type/status/manager.

Toolbar and row actions: Add Team, Edit, Manage Members, Deactivate.

Create/edit form: Team name, description, type (Sales/Support/Account/General), manager, members, queue mapping.

Detail view/tabs: Settings pages: lead sources/statuses, pipeline stages/transitions, loss reasons, service categories, ticket categories/priorities/statuses, business hours, holidays, SLA policies, notification defaults, file limits.

Validation/business rules: Use tenant configuration tables; validate referenced values; prevent deletion of values in use—deactivate instead.

5. Dashboard and management overview

Business purpose: Give executives and managers a quick, reliable view of sales activity, pipeline health, customer work and support performance.

Primary stakeholders: Business owner, executive, sales manager, support manager, account manager and authorized users.

Expected outcome: Users see KPIs and work requiring attention without exposing data outside their permissions.

AI involvement: AI explanations can be added later as generated interpretation with links to underlying records; KPI numbers remain deterministic.

5.1 Role-based dashboard

Purpose: Show relevant metrics and actionable queues.

List/grid columns: Dashboard cards/charts rather than one master grid: new leads, active leads, overdue tasks, open pipeline value, opportunities by stage, won/lost trend, tickets by status/priority, SLA at risk, recent activities.

Filters/search: Date range, team, owner, pipeline, ticket queue, account segment; filters only affect authorized data.

Toolbar and row actions: Refresh, date-range selector, drill-through links, Export Excel for supported tabular widgets only.

Create/edit form: Dashboard personalization: preferred date range, default team, visible widgets. No arbitrary SQL/report builder in MVP.

Detail view/tabs: Widgets: KPI cards; lead funnel; pipeline by stage; opportunity ageing; upcoming/overdue activities; ticket workload; SLA performance; recent changes.

Validation/business rules: Every KPI has a documented definition and date basis. Display last refreshed time. Do not compare currencies without normalization.

6. Lead management

Business purpose: Capture and qualify prospective business interest, assign ownership, track follow-ups and convert qualified prospects into linked account/contact/opportunity records.

Primary stakeholders: Sales executive, sales manager, organization administrator.

Expected outcome: Every active lead has an owner, status and next action; duplicates are reviewed rather than silently merged.

AI involvement: AI can summarize a lead, suggest missing qualification questions and draft a follow-up email. Predictive lead scoring is deferred until sufficient historical conversion data exists.

6.1 Lead list

Purpose: Search, filter, prioritize and manage leads.

List/grid columns: Lead No., lead/company name, contact name, email, phone, source, status, rating/qualification, owner, team, created date, last activity, next action/due date.

Filters/search: Keyword (name/company/email/phone); status; source; owner; team; rating; created date range; last activity range; has overdue task; duplicate flag.

Toolbar and row actions: Add Lead, Import Excel/CSV if approved (CSV import is not an export format), Export Excel, View, Edit, Assign, Convert, Archive/Delete, Duplicate Review.

Create/edit form: Company/person type; company name; first/last name; title; email; phone; website; industry; source; source detail; country/region; need/description; budget range optional; authority/contact role; expected timing; owner; team; status; rating; next action/due date; tags; notes.

Detail view/tabs: Tabs: Overview, Qualification, Activities, Notes, Related opportunities (after conversion), Audit. Duplicate candidates compare side-by-side.

Validation/business rules: Required fields configurable; normalized email/phone; duplicate warning by email/domain/company; conversion requires mapping and user confirmation.

6.2 Lead detail / edit

Purpose: Maintain full qualification context and history.

List/grid columns: Detail header: Lead No., company/contact, status, owner, rating, source, next action.

Filters/search: Activity date/type, subject, owner, due date, outcome.

Toolbar and row actions: Edit, Assign, Add Task, Log Call/Meeting, Convert, Disqualify, Archive, AI Summary, Draft Follow-up.

Create/edit form: Same lead fields as create form plus disqualification reason and qualification notes.

Detail view/tabs: Activity timeline, qualification checklist, duplicate candidates, AI insight panel with evidence.

Validation/business rules: AI-generated content labeled as draft; never send external communication automatically.

6.3 Lead conversion wizard

Purpose: Convert a qualified lead into linked account, contact and optionally opportunity.

List/grid columns: Preview: account match/new account, contact match/new contact, opportunity create/skip, owner mapping.

Filters/search: Search existing account/contact; show likely duplicates.

Toolbar and row actions: Convert (confirmation), Cancel; authorized merge path only after compare.

Create/edit form: Step 1 choose existing/new account; step 2 choose existing/new contact; step 3 optional opportunity details; step 4 review.

Detail view/tabs: Preserve original lead, notes and activities; link them to created records where applicable; record conversion actor/time.

Validation/business rules: Prevent duplicate account/contact creation without explicit decision; conversion transaction must be atomic.

7. Account and contact management

Business purpose: Maintain the organization-level relationship and the people involved in buying, delivery and support.

Primary stakeholders: Sales, account management, support (limited), organization administrator.

Expected outcome: One customer 360 record connects contacts, leads, opportunities, proposals, activities, tickets and shared documents.

AI involvement: AI can generate a Customer 360 summary from records the user may access and highlight evidence-based engagement/support risks.

7.1 Account list

Purpose: Manage prospect and customer organizations.

List/grid columns: Account No., organization name, relationship status, industry, website, primary contact, owner, account manager, country/city, open opportunities count, open tickets count, last interaction, created date.

Filters/search: Keyword; relationship status (Prospect/Customer/Inactive Customer/Former Customer); industry; owner/account manager; geography; created date; last interaction; has open ticket; has open opportunity.

Toolbar and row actions: Add Account, Export Excel, View, Edit, Assign Owner, Add Contact, Add Opportunity, Merge Duplicate, Archive/Delete.

Create/edit form: Organization name, website, industry, company size optional, phone, email, address, country, relationship status, owner, account manager, account source, service interests, notes, tags.

Detail view/tabs: Tabs: Overview, Contacts, Opportunities, Proposals, Activities, Tickets (permission filtered), Documents, Timeline, Notes, Audit.

Validation/business rules: Account name required; status transitions controlled; merging preserves links/history and requires compare + approval.

7.2 Account detail / Customer 360

Purpose: Present a chronological, permission-filtered relationship view.

List/grid columns: Timeline event date, type, subject, actor, related record, visibility, summary.

Filters/search: Date range, event type, owner, visibility; account status and owner filters on list.

Toolbar and row actions: Edit Account, Add Contact, Log Activity, Create Opportunity, Create Ticket (permission), Customer 360 AI Summary.

Create/edit form: Account form plus relationship review date, service interests, relationship notes and health flags (manual/evidence-based).

Detail view/tabs: Overview KPI strip; key contacts; open opportunities; proposals; open tickets; recent activities; shared documents; relationship risks.

Validation/business rules: AI summary must state coverage date range and link factual claims to underlying records; exclude inaccessible records.

7.3 Contact list and detail

Purpose: Manage people associated with customer/prospect organizations.

List/grid columns: Name, account, job title, email, phone, contact role, primary flag, owner, status, last interaction.

Filters/search: Keyword; account; role (decision-maker/technical/commercial/billing/primary); owner; status; last interaction.

Toolbar and row actions: Add Contact, Export Excel, View, Edit, Set Primary, Link/Unlink Account, Archive/Delete.

Create/edit form: Account, first/last name, title, email, phone/mobile, department, role, primary flag, preferred contact method, owner, notes, consent/communication preference where required.

Detail view/tabs: Tabs: Overview, Activities, Opportunities, Tickets (permitted), Notes, Audit.

Validation/business rules: Email normalized; account association required unless standalone lead contact; primary-contact uniqueness rules defined per account/role.

8. Opportunity and sales pipeline

Business purpose: Track each potential commercial engagement from discovery through proposal, negotiation and final outcome.

Primary stakeholders: Sales executive, sales manager, account manager and administrator.

Expected outcome: Managers can inspect pipeline value by stage, owner and close period; every stage change is traceable.

AI involvement: AI can summarize deal history and recommend next actions with evidence. Win-probability ML is deferred until representative historical outcomes exist.

8.1 Opportunity list

Purpose: Search and manage deals.

List/grid columns: Opportunity No., name, account, primary contact, service type, stage, estimated value, currency, probability (configured/manual only), expected close date, owner, last activity, next action, stale flag.

Filters/search: Keyword; account; stage; owner/team; service type; value range; currency; expected close date range; created range; stale/inactive threshold; next action overdue.

Toolbar and row actions: Add Opportunity, Export Excel, View, Edit, Change Stage, Add Activity, Create Proposal, Mark Won, Mark Lost, Archive.

Create/edit form: Account, name, service type, description, primary contact, owner/team, stage, estimated value, currency, expected close date, needs, budget, decision criteria, competitors, decision process, next steps.

Detail view/tabs: Tabs: Overview, Stage History, Activities, Proposals, Contacts, Notes, AI Review, Audit.

Validation/business rules: Won requires outcome date and final value. Lost requires configured loss reason and optional notes. Stage transitions must follow configured graph.

8.2 Pipeline board

Purpose: Visualize opportunities grouped by configured stage.

List/grid columns: Cards show opportunity name, account, value/currency, close date, owner, stale indicator, next action.

Filters/search: Pipeline, owner/team, close period, service type, value band.

Toolbar and row actions: Drag/drop stage change only if authorized; otherwise use Change Stage dialog. Export Excel uses current filters.

Create/edit form: Stage-change dialog requires reason optionally and mandatory reason for configured transitions.

Detail view/tabs: Board columns use tenant-defined order and stage color; provide accessible alternative list/table view.

Validation/business rules: Drag/drop is not an authorization bypass; server validates allowed transition and logs old/new stage, actor and reason.

8.3 Opportunity detail

Purpose: Show complete deal context and next best action.

List/grid columns: Stage history grid: from, to, changed at, changed by, reason. Related proposal versions and activities.

Filters/search: Date/type filters on timeline.

Toolbar and row actions: Edit, Change Stage, Add Task/Meeting, Create Proposal, Mark Won/Lost, AI Opportunity Review.

Create/edit form: Opportunity form; outcome fields shown only for Won/Lost.

Detail view/tabs: Overview, Qualification, Stage History, Activities, Proposals, Customer contacts, AI review, Audit.

Validation/business rules: Stale definition configurable; risk indicators cite facts such as no activity, close-date slippage or missing decision-maker.

9. Activities, tasks, meetings and communication history

Business purpose: Make customer-facing work visible and preserve interaction history for leads, accounts, contacts, opportunities and tickets.

Primary stakeholders: All internal business users; managers for team workload.

Expected outcome: Users see due-today, upcoming and overdue work; interactions appear in linked customer timelines.

AI involvement: AI can convert supplied meeting notes/permitted transcripts into summary, decisions and action items, and draft a follow-up message for review.

9.1 Activity/task list and My Work

Purpose: Manage assigned work and interactions.

List/grid columns: Task No., subject, type, linked record, account/contact, owner, due date/time, priority, status, created by, overdue indicator.

Filters/search: Keyword; type; status; priority; owner; due date range; overdue; linked module/account; created by.

Toolbar and row actions: Add Task, Log Call, Schedule Meeting, Export Excel, View, Edit, Complete, Reassign, Cancel.

Create/edit form: Subject, description, type (task/call/meeting/demo/email/other), related record type/id, account/contact, owner, due/start/end date, timezone, priority, status, reminder, participants, outcome, notes.

Detail view/tabs: Views: My Work, Due Today, Upcoming, Overdue, Team Workload, Calendar (optional), Activity Timeline.

Validation/business rules: Completed tasks retain completion time/user; reminders respect timezone/preferences; link target must be accessible.

9.2 Meeting / interaction detail

Purpose: Record discussion and follow-up commitments.

List/grid columns: Date/time, type, subject, account, participants, owner, outcome, action items, linked tasks.

Filters/search: Date range, type, owner, account, participant.

Toolbar and row actions: Add Interaction, Edit, Create Follow-up Task, AI Summarize Notes, Draft Follow-up.

Create/edit form: Notes, decisions, action items, participants, outcomes, attachments and transcript only where lawfully provided.

Detail view/tabs: Timeline card plus structured decisions/action items.

Validation/business rules: AI uses only user-provided or permitted transcript; review before saving/sending generated content.

10. Proposal and quotation management

Business purpose: Prepare consistent commercial proposals linked to opportunities and preserve revisions, approvals and customer response history.

Primary stakeholders: Sales executive, sales manager, account manager and administrator.

Expected outcome: Each proposal has a version history and a clearly identifiable approved/sent version.

AI involvement: AI may create an editable draft and completeness check; it must not invent prices, delivery dates or alter approved legal/commercial wording.

10.1 Proposal list

Purpose: Track proposals and their approval/sending lifecycle.

List/grid columns: Proposal No., title, account, opportunity, version, amount/currency, status, owner, approver, created date, valid until, sent date, response.

Filters/search: Keyword; account; opportunity; status; owner; approver; amount range; created/sent/expiry date range.

Toolbar and row actions: Create Proposal, Export Excel, View, Edit Draft, Create New Version, Submit for Review, Approve/Reject, Record Sent/Response, Archive.

Create/edit form: Opportunity/account, title, scope summary, deliverables, assumptions, exclusions, timeline, commercial amount, currency, payment terms, validity date, template, owner.

Detail view/tabs: Tabs: Content, Versions, Approval History, Customer Response, Related Activities, Audit.

Validation/business rules: Only draft/in-review versions editable; approved/sent version immutable—changes create a new version.

10.2 Proposal editor and approval

Purpose: Compose structured proposal content and route approvals.

List/grid columns: Sections: executive summary, scope, deliverables, assumptions, exclusions, timeline, pricing, payment terms, validity, standard terms.

Filters/search: Template selector; version status.

Toolbar and row actions: Save Draft, Preview, Submit for Approval, Approve, Reject with reason, Record Sent, Record Accepted/Rejected.

Create/edit form: Rich text/structured section editor; price lines only if line-item pricing is approved as a product requirement; do not infer tax/invoice behavior.

Detail view/tabs: Version compare, reviewer comments, approval decision and send record.

Validation/business rules: Approval policy configurable by amount/discount/role. Customer email/send integration must be explicitly configured; no automatic sending from AI.

11. Customer support and ticket management

Business purpose: Capture, prioritize, assign, track and resolve post-sale customer issues and requests.

Primary stakeholders: Customer portal user, support agent, support manager, account manager and administrator.

Expected outcome: Every ticket has a reference, account, owner/queue, priority, status, public conversation and internal notes; SLA performance is measurable.

AI involvement: AI can suggest category/priority, summarize long threads, retrieve approved knowledge and similar resolved tickets, and draft a cited customer reply. Humans remain responsible for diagnosis and external communication.

11.1 Ticket queue / list

Purpose: Operate support workload across queues.

List/grid columns: Ticket No., subject, account, contact/requester, category, priority, status, queue, assignee, created date, last reply, first response due, resolution due, SLA state, age.

Filters/search: Keyword; status; priority; category; queue; assignee; account; created date; SLA state (on track/at risk/breached); unassigned; reopened.

Toolbar and row actions: Create Ticket, Export Excel, View, Assign/Reassign, Escalate, Add Public Reply, Add Internal Note, Resolve, Reopen, Merge Related (review).

Create/edit form: Subject, description, account, requester/contact, category, priority, attachments, source (portal/internal/email integration later), queue, assignee.

Detail view/tabs: Views: My Tickets, Queue, Unassigned, At Risk, Breached, Waiting for Customer, Recently Resolved.

Validation/business rules: SLA calculations use configured business hours, holidays and pause rules. Internal notes must never appear in portal/public API.

11.2 Ticket detail / workspace

Purpose: Give agents a complete issue workspace.

List/grid columns: Conversation: timestamp, author, public/internal visibility, message, attachment list. Related ticket grid: ticket no., subject, status, category, similarity/reason.

Filters/search: Conversation date range; related ticket status/category.

Toolbar and row actions: Reply Public, Add Internal Note, Assign, Escalate, Change Priority, Change Status, Resolve, Reopen, AI Summarize, Suggest Classification, Search Knowledge, Draft Reply.

Create/edit form: Reply composer with explicit Public/Internal selector; resolution code and resolution summary required on resolve; reassignment reason optional/configured.

Detail view/tabs: Header SLA timers; requester/account context; conversation; internal notes; attachments; similar tickets; linked knowledge; audit history.

Validation/business rules: Every public reply and internal note has distinct visibility flag and separate permission; attachments scanned and access-checked.

11.3 SLA configuration and monitoring

Purpose: Define response/resolution targets and identify at-risk/breached tickets.

List/grid columns: Policy name, priority/category scope, business calendar, first response target, resolution target, pause statuses, active flag.

Filters/search: Policy, priority, category, active.

Toolbar and row actions: Add/Edit/Deactivate policy; SLA dashboard; Export Excel.

Create/edit form: Business hours/timezone, holidays, priority target durations, pause rules (e.g. Waiting for Customer), escalation threshold.

Detail view/tabs: Ticket SLA timeline and breach history.

Validation/business rules: Define first response as first public human response; automated acknowledgement excluded unless policy explicitly says otherwise. Resolution time pauses per approved rules.

12. Customer portal

Business purpose: Provide a limited, secure external experience for customer organization users to submit and monitor support requests.

Primary stakeholders: Customer portal users and internal portal administrators/support agents.

Expected outcome: External users see only their own organization's permitted tickets, responses, attachments and published/shared knowledge.

AI involvement: Customer-facing AI is not in initial scope; employee-facing AI must not expose internal-only notes or documents.

12.1 Portal home

Purpose: Show the customer's support snapshot.

List/grid columns: Summary cards: open tickets, waiting for customer, recently resolved; ticket list with reference, subject, status, priority, last update.

Filters/search: Ticket status, date range, keyword.

Toolbar and row actions: New Ticket, View Ticket, Search Published Knowledge, Profile/Sign Out.

Create/edit form: Portal profile: name, email, phone, notification preferences. Ticket creation fields match support intake.

Detail view/tabs: Home, My Tickets, Knowledge Base, Profile, Help.

Validation/business rules: All queries are scoped to portal user's tenant/account and permitted contact/organization policy.

12.2 Portal ticket detail

Purpose: Allow customer to view and reply to permitted tickets.

List/grid columns: Ticket reference, subject, status, priority, created/updated, public conversation, customer-visible attachments.

Filters/search: Conversation date range.

Toolbar and row actions: Add Reply, Upload Attachment, Reopen (only if configured), Return to Ticket List.

Create/edit form: Public reply text and attachments only.

Detail view/tabs: Never return internal notes, internal-only attachments, internal assignee notes or commercial account details.

13. Knowledge base and document management

Business purpose: Create and govern reusable company knowledge that employees can search and AI can use as a trusted retrieval source.

Primary stakeholders: Knowledge administrator, support manager, support agents, sales users and permitted internal users.

Expected outcome: Published content has ownership, versioning, classification and a traceable lifecycle; unauthorized/draft/archived content is excluded from general AI retrieval.

AI involvement: Foundation for RAG: source title and relevant section/page must be returned with answers.

13.1 Knowledge article list

Purpose: Find and govern knowledge articles.

List/grid columns: Article ID, title, category, tags, owner, status, visibility classification, version, published date, updated date, ingestion/index status.

Filters/search: Keyword; category; tags; owner; status; visibility; updated/published date; indexing status.

Toolbar and row actions: Add Article, Upload Document, Export Excel (metadata only), View, Edit Draft, Submit Review, Publish, Archive, Supersede, Reindex.

Create/edit form: Title, category, content, tags, owner, status (Draft/In Review/Published/Archived/Superseded), visibility (organization-wide/team-restricted/customer-specific), effective date, source file, version notes.

Detail view/tabs: Tabs: Content, Versions, Access, Usage/Feedback, Ingestion Status, Audit.

Validation/business rules: Only authorized publisher can publish; publication/withdrawal triggers re-indexing; archived/superseded versions excluded from default retrieval.

13.2 Document upload and processing

Purpose: Upload PDF, DOCX and TXT documents within configured limits and track ingestion.

List/grid columns: File name, type, size, uploader, upload date, classification, status, page count, chunks, embedding model/version, last indexed.

Filters/search: File name; type; uploader; date; status; classification; processing error.

Toolbar and row actions: Upload, Retry, Cancel, View Extracted Text (permission), Reindex, Archive/Delete.

Create/edit form: File, title, category, owner, visibility, allowed groups/accounts, effective date, language, retention metadata.

Detail view/tabs: Pipeline steps: uploaded → malware/type validation → extraction → normalization → chunking → metadata/ACL assignment → embedding → vector/index persistence → ready/failed.

Validation/business rules: Use asynchronous worker; show per-document state and retry; never mark ready before all required chunks are persisted. Enforce file size/type limits.

13.3 Knowledge search

Purpose: Search published, permitted knowledge using keyword and semantic retrieval.

List/grid columns: Result title, snippet, category, source type, updated date, access label, relevance indicator.

Filters/search: Keyword, category, tag, content type, date, visibility scope.

Toolbar and row actions: Search, Open Article, Give Helpful/Not Helpful feedback. No export of document content by default.

Create/edit form: Search query only; advanced filters.

Detail view/tabs: Show source sections/page numbers and article version; AI assistant reuses the same authorized retrieval service.

Validation/business rules: Apply tenant and user ACL filters before context construction; do not expose hidden document existence through result counts.

14. Reporting and analytics

Business purpose: Provide consistent operational and management reporting based on approved metric definitions.

Primary stakeholders: Business owner, sales manager, support manager, account manager and authorized analysts.

Expected outcome: Reports reconcile to source records and honor tenant, role and record-level access.

AI involvement: AI may explain report trends using retrieved report facts; it must not fabricate KPI values.

14.1 Standard reports

Purpose: Provide fixed, documented reports for sales, account and support.

List/grid columns: Lead funnel; lead source; lead response time; opportunities by stage/owner/close period; win/loss; pipeline ageing; activity completion; account engagement; ticket volume; first response; resolution; SLA attainment; reopened tickets.

Filters/search: Date range, team, owner, stage/status, source, account, service type, priority, queue, SLA state; filters vary by report.

Toolbar and row actions: Run, Reset, Save View (optional), Drill Through, Export Excel.

Create/edit form: Report parameter panel; no unrestricted user SQL or formula builder in initial release.

Detail view/tabs: Report definition page includes metric definition, inclusion/exclusion rules, timezone and last refreshed time.

Validation/business rules: First response = first public human response; resolution = resolved timestamp; SLA denominator and pause rules configured. Win rate cohort definition must be documented.

14.2 Export center

Purpose: Track larger or asynchronous Excel exports.

List/grid columns: Export ID, report/module, requested by, requested date, filter summary, row count, status, expiry, download action.

Filters/search: Module, requester, date range, status.

Toolbar and row actions: Request Export, Download, Cancel (if queued).

Create/edit form: Selected fields, current filters, file name; server validates export permission and column allowlist.

Detail view/tabs: Export status queued/processing/ready/failed/expired; audit event on request and download.

Validation/business rules: Protect against formula injection; expire temporary files; never create a public object URL.

15. AI platform – complete technical design

AI is an assistance layer, not a system of record. CRM business rules, permissions, calculations and state transitions remain deterministic application code. LLMs may interpret language, summarize, draft, classify, retrieve and plan within a strict tool allowlist.

AI capability map

Capability	Use in CRMPro AI	Initial priority / guardrail
LLM chat	Natural-language assistant for permitted CRM data and knowledge questions.	Read-only first; tenant/user scoped.
Structured CRM Q&A;	Translate user intent into validated filters and approved application queries.	No arbitrary SQL; query schema allowlist.
RAG knowledge Q&A;	Answer policies/manual/how-to questions from published authorized sources.	Citations required; abstain if unsupported.
Summarization	Lead, account 360, opportunity, ticket thread and meeting notes.	Facts linked to records; date range stated.
Draft generation	Follow-up email, proposal sections, support reply.	Editable draft; human review before sending.
Classification	Ticket category/priority suggestions; lead qualification hints.	Confidence and override; low confidence goes to human.
Agent workflows	Stalled opportunity review, customer summary, ticket analysis, prepare tasks.	Preview + explicit approval for consequential changes.
Embeddings	Represent approved document chunks and query text for semantic search.	Store model/version/dimensions; re-embed on model change.
Reranking	Improve top retrieved chunks using a reranker or model scoring.	Optional after baseline retrieval evaluation.
Predictive ML	Lead scoring / deal win probability.	Could/future only after enough representative labeled outcomes.

AI architecture components

- AI Application Service: orchestrates use cases and owns authorization, limits, prompt assembly and output validation.
- Model Gateway: provider-neutral IChatClient/embedding abstraction; adapters for selected provider(s); model routing by task.
- Prompt Registry: versioned system/task prompts, variables, policy version and test fixtures; no prompt strings scattered across controllers.
- Conversation Service: tenant/user-owned conversations, messages, context window and retention policy.
- Retrieval Service: permission-aware hybrid search over knowledge chunks; returns source IDs, title, page/section, version and snippet.
- Tool Registry: allowlisted typed business tools with JSON schemas, per-tool permission policy, validation, idempotency and audit.
- Agent Orchestrator: bounded workflow state machine; maximum steps, timeouts, token budget and retry policy.
- Approval Service: pending action preview, approver, expiry, approval/rejection reason and one-time execution token.
- Usage & Cost Service: tokens, model, latency, estimated cost, tenant quotas, user quotas and feature switches.
- Evaluation Service: golden datasets, answer/source correctness, tool success, prompt regression and feedback.

Provider and model strategy

Use Microsoft.Extensions.AI as the application-facing abstraction and implement one concrete provider adapter initially. Keep provider-specific configuration isolated. The exact provider/model is an OPEN operational decision based on privacy terms, supported region, price, latency, context size, structured output, embeddings and tool calling. Do not bind Domain/Application code to a vendor SDK.

Task	Model characteristics to select	Fallback
Chat / reasoning	Reliable instruction following, structured output, tool calling, adequate context window.	Show temporary AI unavailable; core CRM remains functional.
Embeddings	Stable dimension, multilingual/business text quality, acceptable cost and provider data terms.	Queue ingestion; keep keyword search available.
Reranking (optional)	Improves top-k relevance on evaluation set.	Use vector + keyword ranking only.
Speech/transcription (future)	Only if user supplies permitted recordings/transcripts and scope is approved.	Manual notes.

AI user experience pages

Page / component	Content and behavior
AI Assistant workspace	Conversation list, new chat, message thread, streaming response, citations, feedback, context/record references, stop/retry, clear context.
AI side panel on records	Contextual actions: summarize lead/account/opportunity/ticket; draft follow-up; analyze stalled deal; suggest next steps.
AI action preview	Action name, target records, proposed field changes/tasks/messages, evidence, warnings, permissions, approval requirement, Approve/Reject/Edit.
AI usage dashboard (admin)	Requests, tokens, estimated cost, model, failures, latency, user/feature breakdown, quota utilization.
Prompt/evaluation admin (restricted)	Prompt versions, active version, evaluation run results, test cases and release approval.

16. RAG – retrieval augmented generation implementation

Knowledge ingestion pipeline

Step	Implementation requirement	Failure behavior
1. Upload	Accept PDF, DOCX, TXT; enforce extension, MIME sniffing, size, malware scan and tenant storage path.	Reject unsafe/unsupported file; do not enqueue.
2. Register	Persist Document, version, tenant, owner, classification, source metadata and ingestion job.	Keep state Failed with reason; user can retry.
3. Extract	Use format-specific parsers; retain page/heading/section metadata where possible.	Mark extraction failure; never embed empty text silently.
4. Normalize	Remove repeated headers/footers/noise; preserve tables and meaningful structure where possible.	Store normalized text and extraction version.
5. Chunk	Split by headings/semantic boundaries with token-aware size and overlap; attach source/page/section metadata.	Record chunker version and chunk counts.
6. ACL metadata	Attach tenant, document, version, publication state, visibility, allowed teams/accounts and effective dates.	Missing ACL metadata means deny retrieval.
7. Embed	Batch embeddings with retry/backoff, provider quota and dimension validation.	Retry transient errors; dead-letter permanent errors.
8. Persist	Store chunk text/metadata/embedding in PostgreSQL pgvector with indexes.	Atomic version activation; do not mix partial versions.
9. Activate	Mark version searchable only when all chunks succeed and document is Published.	Draft/restricted/archived content remains excluded.
10. Reindex/delete	Rebuild on content, ACL, status or embedding model changes; remove stale vectors safely.	Track job status and audit.

Chunk metadata minimum

Field	Purpose
TenantId, DocumentId, DocumentVersionId, ChunkId	Isolation and source identity.
ArticleId (optional), Title, SourceFileName	Human-readable source attribution.
PageNumber, HeadingPath, SectionLabel, Character/Token offsets	Citation precision and traceability.
PublicationStatus, EffectiveFrom/To, IsArchived	Lifecycle eligibility.
VisibilityScope, AllowedTeamIds, AllowedAccountIds, Classification	Authorization filter.
Language, ContentType, Tags, OwnerId	Retrieval filters.
EmbeddingModel, EmbeddingVersion, ChunkerVersion	Reproducibility and re-index control.

Query-time RAG pipeline

- Authenticate caller; resolve tenant, user, role, team, account access and feature entitlements.
- Classify intent (knowledge question vs structured CRM question vs mixed). Do not rely on the LLM to decide authorization.
- Normalize/rewrite query if useful; preserve original query for traceability.
- Generate query embedding; run vector similarity and keyword/full-text search with tenant, publication, effective-date and ACL filters applied in the database query.
- Combine rankings (e.g. reciprocal rank fusion); optionally rerank top candidates. Keep top-k and token budget bounded.
- Validate each source again against current permissions/status before constructing context.
- Build context with source labels and page/section metadata; clearly delimit retrieved text as untrusted data.

- Call LLM with grounded-answer instructions; require claims to map to source IDs where possible.
- Validate citation IDs against retrieved sources; reject fabricated source references. If evidence is insufficient, answer that no reliable source was found.
- Return answer, citations, retrieval metadata, confidence/coverage signal and feedback controls; log model/prompt/retrieval versions without leaking sensitive content.

Critical RAG security rules

- Tenant and ACL filters happen before retrieval results are returned to the model. Post-filtering alone is not acceptable.
- Never index draft, archived, superseded or unauthorized content for general retrieval; re-check status/ACL at query time to handle revocation.
- Treat documents and retrieved chunks as untrusted data. Ignore instructions embedded in source text that attempt to override system policy or call tools.
- Do not expose restricted document titles/snippets or existence via counts, error messages, citations or debug logs.
- Use minimal necessary context; redact sensitive data where appropriate; document provider retention/data processing terms.
- RAG is not permission granting: user access to the underlying record/document remains the source of truth.

Initial RAG defaults – tune by evaluation

Setting	Starting implementation choice
Chunking	Heading-aware/token-aware chunks; start around 500–800 tokens with 80–120 token overlap; evaluate by document type.
Retrieval	Hybrid: PostgreSQL full-text + pgvector similarity; initial top-k 8–12, then context selection.
Reranking	Optional; introduce only if a reviewed evaluation set shows measurable improvement.
Abstention	If no sufficiently relevant authorized source, state no reliable source found; do not invent.
Citations	Article/document title + version + page/section; citation opens authorized source view.
Index lifecycle	New version staged; activate only when complete; old version deactivated atomically.

17. AI assistant, structured CRM Q&A; and agents

Structured CRM questions

Examples: “Show open opportunities above 50,000 closing this quarter” or “Which tickets are at risk of breaching SLA?” The assistant must map intent to a finite set of typed query tools, validate parameters, call Application queries and present results. Do not generate or execute arbitrary SQL.

Allowed tool example	Input schema concept	Output
SearchLeads	status, ownerId, source, createdFrom/To, text, paging	Authorized lead summaries + record links.
SearchOpportunities	stageIds, ownerId/teamId, minValue, currency, closeFrom/To, staleOnly	Authorized opportunity rows + totals with currency grouping.
GetAccountSummary	accountId, dateRange	Permission-filtered Customer 360 facts.
SearchTickets	status, priority, queueId, assigneeId, slaState, dateRange	Authorized ticket rows.
SearchKnowledge	query, category, tags	Authorized published source references.
GetReportMetric	metricKey, filters, dateRange	Deterministic KPI value + definition + timestamp.

Tool calling contract

- Every tool has a stable name, JSON input schema, output DTO, required permission, tenant context and timeout.
- Validate arguments server-side (allowlisted enum values, bounded dates/page size, IDs scoped to tenant).
- Tool handlers call Application services, never raw DbContext or arbitrary SQL from the model.
- Tool output is minimized; do not return secrets, private notes or fields the caller cannot read.
- Read-only tools may run without approval if user has permission. Mutating tools create a preview/pending action.
- All tool invocations record actor, tenant, tool name/version, arguments redacted as needed, result status, correlation id and latency.

Agent workflow definitions

Workflow	Steps	Approval / guardrail
Customer 360 summary	Collect permitted account, contact, opportunity, proposal, activity and ticket facts; summarize with links and date coverage.	Read-only; citations/record links mandatory.
Stalled opportunity review	Find open opportunities with no meaningful activity beyond configured threshold; summarize evidence; suggest next action.	Read-only analysis; proposed task creation requires preview and approval.
Ticket analysis	Summarize reported issue, attempted steps, open questions; retrieve approved knowledge and similar resolved tickets.	No automatic technical fix; unsupported knowledge yields abstention.
Prepare follow-up tasks	Identify stale deals and draft task title/owner/due date/linked opportunity.	Preview all tasks; verify assignee permission; explicit approval before create.
Draft customer reply	Use ticket thread and approved KB sources to create a draft.	Never send automatically; user edits and explicitly sends through normal workflow.

Agent state machine

Draft → Validating → AwaitingApproval → Approved → Executing → Completed / PartiallyCompleted / Failed / Rejected / Expired. Persist each transition, tool result, idempotency key and approver.

Approval UX and execution controls

- Preview exact target records and field changes/actions before approval; display external visibility and irreversible effects.
- Approver must have both the action permission and approval permission. The requester cannot self-approve when policy prohibits it.

- Approval token is single-use, scoped to exact payload hash, expires and is invalidated if target record changed.
- Use idempotency keys to prevent duplicate tasks/messages on retries. Revalidate permissions and business rules at execution time.
- Partial failures are visible; allow safe retry only for failed idempotent steps. Record user corrections and final outcome.
- Actions that send external messages, materially change records or affect customer commitments require authorized human approval.

AI prompt and output controls

- Separate system policy, task prompt, tenant configuration and untrusted retrieved/user content.
- Version prompts; include owner, purpose, variables, model compatibility, change reason and evaluation results.
- Use JSON schema/structured output for classifications, summaries with typed sections and tool arguments.
- Validate output server-side; never trust model-generated IDs, URLs, SQL, HTML, file paths or permissions.
- Apply prompt injection defenses, output sanitization, token/time limits, tool step limits, rate limits and provider retries.
- Stream tokens to UI where useful, but do not stream unvalidated tool actions as if completed.

18. AI evaluation, monitoring and governance

Evaluation area	Test / metric	Release gate
RAG retrieval	Recall@k / source relevance on reviewed question-source pairs.	Critical questions retrieve authorized source; unauthorized sources never appear.
Grounded answer	Key claim support, citation correctness, abstention quality.	No fabricated citation; unsupported answer abstains.
Structured Q&A;	Filter correctness, result parity against deterministic query, permission correctness.	No arbitrary query; results reconcile with CRM UI/report.
Summarization	Factual consistency, omitted critical facts, date coverage, source links.	No invented dates/values; source records accessible.
Classification	Precision/recall by category/priority, confidence calibration, override rate.	Human override always possible; low-confidence manual queue.
Agent workflow	Task success, partial completion, duplicate prevention, user correction.	No unapproved consequential action; retries idempotent.
Safety	Prompt injection, cross-tenant leakage, data exfiltration, malicious attachments.	Zero cross-tenant and unauthorized record retrieval in regression suite.
Operations	Latency, timeout, provider error, token/cost, quota, queue failure.	Budget thresholds and graceful fallback work.

Required AI observability

- Trace ID, tenant ID, user ID (pseudonymize in analytics where possible), feature, model/provider, prompt version, retrieval version, embedding version.
- Input/output token counts, latency, estimated cost, retries, rate-limit events, tool calls, approval outcomes and errors.
- Do not log full sensitive prompts/documents by default. Define retention, redaction and access controls for AI conversation logs.
- Tenant-level quotas and feature switches; alert on spend spikes, repeated failures, unusually large retrievals and abnormal tool activity.
- Track user feedback (helpful/not helpful, accepted/edited/rejected) separately from factual quality evaluation.

Predictive ML – deliberately later

Lead scoring and opportunity win probability are Could requirements. Do not ship scores merely to demonstrate ML. First establish reliable historical outcomes, sufficient volume, stable definitions, representative cohorts and a reviewed evaluation dataset. If data is insufficient, keep the feature disabled and document why.

- Data readiness: labels, timestamped snapshots, no leakage from post-outcome fields, missing-value strategy and cohort definition.
- Baseline: compare a simple interpretable baseline with LightGBM or another tabular model; holdout by time, not random split alone.
- Evaluation: calibration, precision/recall at operational threshold, lift, subgroup performance and drift.
- Deployment: model registry/version, feature schema, inference endpoint, explanation, fallback to no score, monitoring and retraining governance.
- Never let a predictive score autonomously reject a lead, change deal pricing or send customer communications.

19. Database and Code First data model

Use EF Core Code First. Entities and configurations are source-controlled; migrations are reviewed, deterministic and applied through deployment automation. Avoid database-first scaffolding.

Core entity inventory (logical baseline)

Module	Entities / aggregate roots	Important relationships / notes
Tenant	Tenant, TenantSetting, TenantFeature, TenantUsageQuota	Root scope for all tenant-owned records.
Identity	ApplicationUser, Role, Permission, UserRole, RolePermission, Team, TeamMember, RefreshTokenSession, Invitation	Identity may use ASP.NET Core Identity tables; tenant association explicit.
Lead	Lead, LeadSource, LeadStatus, LeadAssignmentHistory, LeadDuplicateCandidate, LeadConversion	Conversion preserves original lead and linked history.
Accounts	Account, Contact, AccountContactRole, AccountMergeHistory, AccountNote	Account has many contacts/opportunities/tickets/activities.
Sales	Opportunity, Pipeline, PipelineStage, StageTransitionRule, OpportunityStageHistory, LossReason	Won/Lost outcomes validated; currency explicit.
Activities	Activity, ActivityParticipant, ActivityReminder, ActivityLink, CommunicationEntry	Polymorphic link should use controlled link table, not arbitrary FK string.
Proposals	Proposal, ProposalVersion, ProposalSection, ProposalLine (if approved), ProposalApproval, ProposalTemplate, ProposalSendRecord	Sent/approved version immutable.
Support	Ticket, TicketMessage, TicketAttachment, TicketAssignmentHistory, TicketQueue, TicketCategory, TicketPriority, TicketStatus, SlaPolicy, TicketSlaClock, ResolutionCode, RelatedTicket	Internal/public visibility explicit.
Knowledge	KnowledgeArticle, KnowledgeArticleVersion, KnowledgeDocument, DocumentAccessRule, DocumentIngestionJob, DocumentChunk, Embedding metadata	Vector column in chunk table or dedicated vector table.
AI	AiConversation, AiMessage, AiRun, AiToolInvocation, AiPendingAction, AiApproval, AiPrompt, AiPromptVersion, AiModelConfig, AiUsageRecord, AiEvaluationCase, AiEvaluationRun, AiFeedback	Sensitive retention controls; avoid raw prompt logging by default.
Platform	AuditEvent, Notification, NotificationPreference, OutboxMessage, InboxMessage, ExportJob, FileObject	Audit and integration events are durable.

Code First standards

- Use explicit IEntityTypeConfiguration per entity; keep DbContext lean and module-aware.
- Define primary keys as UUID; use UTC timestamps; use decimal for monetary values (never float/double); store currency ISO code separately.
- Use unique indexes scoped by TenantId (e.g. tenant + normalized email where business rules require). Add indexes for common filters and foreign keys.
- Use optimistic concurrency (xmin mapping or explicit concurrency token) for editable records; return 409 Conflict with a user-readable message.
- Use check constraints for valid ranges and invariant status data where appropriate; enforce complex transitions in domain/application services.
- Use value objects for Money, EmailAddress, PhoneNumber or Address where they add real invariant value; avoid overengineering.
- Use explicit delete behavior; prevent cascading deletion of business history. Prefer archive/soft-delete.

- Keep vector dimensions consistent with selected embedding model; store model identifier and dimension; reindex on model migration.
- Do not run migrations automatically in production API startup. Generate migration script/bundle and apply as controlled deployment step.

Example tenant-scoped base fields

```
Id UUID PK
TenantId UUID NOT NULL (tenant-owned tables)
CreatedAtUtc TIMESTAMPTZ NOT NULL
CreatedByUserId UUID NULL
UpdatedAtUtc TIMESTAMPTZ NULL
UpdatedByUserId UUID NULL
IsArchived BOOLEAN NOT NULL DEFAULT FALSE
ArchivedAtUtc TIMESTAMPTZ NULL
ConcurrencyToken / xmin
```

Important schema decisions

- Tenant and platform/global tables must be clearly separated. Never make TenantId nullable as a shortcut for shared/global data.
- Model configurable statuses/categories as tenant-owned reference tables rather than C# enums where admins must configure them.
- Use enums only for stable technical states; business pipeline stages and ticket statuses are tenant-configurable.
- Store attachments in object storage; DB stores metadata, tenant, object key, size, checksum, MIME type, scan state and access classification.
- Store AI conversation messages separately from business record notes; define retention/export/delete behavior.
- Use PostgreSQL full-text search and pgvector with appropriate indexes; benchmark query plans with realistic tenant datasets.

20. API design and integration standards

API conventions

- Version APIs under /api/v1. Use REST resource routes and consistent nouns.
- Use DTOs, never serialize EF entities. Use separate request and response models.
- List endpoints support page, pageSize, sortBy, sortDirection and validated filters; cap pageSize.
- Use RFC 7807 ProblemDetails for errors with stable error codes and correlationId.
- Return 201 + Location for create, 204 for successful delete/archive where suitable, 409 for concurrency/business conflicts, 403 for forbidden, 404 for inaccessible/nonexistent records according to anti-enumeration policy.
- Use idempotency keys for external/agent-triggered mutations and retryable integrations.
- OpenAPI documents auth scopes, request/response schemas, validation and examples. Angular clients use typed interfaces/services.
- Do not expose internal stack traces, provider keys, SQL or other tenant data in error responses.

Suggested API route inventory

Area	Example routes
Auth	POST /api/v1/auth/login, /refresh, /logout; GET /me; POST /password/forgot, /reset.
Tenant/settings	GET/PATCH /api/v1/organization; GET/PUT /settings; GET/PUT /business-hours.
Users/roles	GET/POST /users; GET/PATCH /users/{id}; POST /users/{id}/invite; GET/POST /roles; PUT /roles/{id}/permissions.
Leads	GET/POST /leads; GET/PATCH /leads/{id}; POST /leads/{id}/assign; /convert; /duplicate-review.
Accounts/contacts	GET/POST /accounts; GET/PATCH /accounts/{id}; /timeline; /merge; GET/POST /contacts.
Opportunities	GET/POST /opportunities; PATCH /opportunities/{id}; POST /{id}/stage; /mark-won; /mark-lost; GET /pipeline.
Activities	GET/POST /activities; PATCH /activities/{id}; POST /{id}/complete; GET /activities/my-work.
Proposals	GET/POST /proposals; POST /{id}/versions; /submit-review; /approve; /reject; /record-sent; /record-response.
Tickets	GET/POST /tickets; GET/PATCH /tickets/{id}; POST /{id}/assign; /messages; /internal-notes; /resolve; /reopen; /escalate.
Knowledge	GET/POST /knowledge/articles; /publish; /archive; /documents; /documents/{id}/reindex; GET /knowledge/search.
Reports	GET /reports/{reportKey}; POST /exports; GET /exports/{id}; GET /exports/{id}/download.
AI	POST /ai/conversations; GET /ai/conversations; POST /ai/conversations/{id}/messages; POST /ai/actions/{id}/approve; /reject; GET /ai/usage.
Audit	GET /audit-events with permission and tenant filters.

Integrations

Integration	Purpose / contract
Email provider (future/optional)	Transactional invites/reminders and approved external communication. Provider adapter; delivery status and retry; never send AI draft without explicit send action.
Calendar (future/optional)	Meeting scheduling/sync only after scope approval; OAuth tokens encrypted; conflict handling.

Integration	Purpose / contract
File/object storage	Pre-signed upload/download URLs short-lived; server verifies object ownership, content type and tenant.
AI provider	Provider gateway; data-processing settings; retries/timeouts; token/cost telemetry; secrets in secret manager.
RabbitMQ	Outbox-published events; durable queues, dead-letter queues, retry policy and idempotent consumers.
Monitoring	OpenTelemetry traces/metrics/logs; alerting for API errors, queue lag, AI spend, SLA scheduler and DB saturation.

21. Frontend implementation specification

Angular application structure

```
src/app/
core/ (auth, guards, interceptors, tenant context, layout, error handling)
shared/ (buttons, dialogs, grids, pipes, validators, empty/loading states)
api/ (typed API clients, models, pagination/filter contracts)
features/
dashboard/ leads/ accounts/ contacts/ opportunities/ activities/
proposals/ tickets/ knowledge/ reports/ ai-assistant/ settings/
layouts/ (authenticated shell, portal shell)
app.routes.ts
```

Reusable frontend building blocks

- DataTablePage: server-side paging/sort/filter, saved column preferences, selection and Excel export trigger.
- FilterPanel: typed filter definitions, apply/reset, URL query state for shareable list views where safe.
- RecordHeader: title, identifier, status badge, owner, primary actions and breadcrumbs.
- FormDialog/Page: typed Reactive Form, async validation, server error mapping, unsaved-change guard.
- ConfirmActionDialog: impact summary, destructive-action confirmation and required reason.
- Timeline: chronological events with visibility labels and related-record links.
- FileUpload: allowed types/size, upload progress, scan status and secure download.
- AIResponsePanel: generated/suggested label, streaming, citations, feedback, copy, retry and report issue.
- PermissionDirective/Guard: UX convenience only; backend remains authoritative.
- Toast/error handler: consistent success, validation, conflict, forbidden, network and provider outage messaging.

Excel export requirements

- Only .xlsx is offered for user-facing exports. Use a maintained .NET library with licensing reviewed for commercial/portfolio use.
- Export current filters by default; make 'selected rows' an explicit option. Include a generation timestamp, tenant label and filter summary in workbook metadata or first sheet.
- Server re-runs the query under current tenant/record permissions; never trust client-supplied row IDs alone.
- Audit export requested/downloaded, user, tenant, module, filters, row count and file expiry.
- Prevent spreadsheet formula injection by escaping cells beginning with =, +, -, @ or control characters.
- Large exports run asynchronously; status appears in Export Center; download URL expires.

Import policy

The BRD calls for basic import/export. If import is included, use a separate controlled import workflow with downloadable Excel template or CSV input only if approved. Validate every row, show preview and row-level errors, reject cross-tenant references, and audit the import. The requested user-facing export format remains Excel only.

22. Security, privacy, reliability and operations

Security controls

- TLS everywhere; secure cookies where applicable; password hashing via ASP.NET Core Identity; MFA is a recommended later control for admins.
- Short-lived access tokens and rotating refresh tokens; revoke on logout, password reset, user deactivation or suspicious session.
- Rate-limit login, password reset, public portal endpoints, AI endpoints and export creation.
- Validate and encode all user input; use parameterized EF queries; avoid dynamic SQL; protect against XSS, CSRF where cookie auth is used, SSRF and path traversal.
- File upload: extension allowlist + MIME sniffing + size cap + malware scanning + random object key + quarantine until clean.
- Secrets only in environment/secret manager; never commit appsettings secrets, provider keys or JWT signing keys.
- Audit sensitive operations: permission changes, exports, merge/delete, proposal approvals, public replies, ticket assignment, AI action approvals/execution and tenant access.

Privacy and AI data governance

- Document exactly what data is sent to each AI provider and for which feature. Minimize context and avoid sending unrelated customer fields.
- Configure provider data retention/training controls according to contractual terms; do not promise zero retention unless verified.
- Define retention periods for CRM records, files, email content, AI messages, embeddings, audit logs and export files.
- Support deletion/withdrawal workflows that remove source documents and derived chunks/embeddings when legally and operationally required.
- Restrict access to AI conversation history; user/tenant scope must be enforced on every read.
- Provide tenant feature controls and usage budgets; disable AI per tenant without affecting core CRM.

Background processing and reliability

Workflow	Queue / job behavior
Document ingestion	Queued, retryable, per-step status, idempotent by document version; dead-letter permanent failure.
Embedding batches	Rate-limited, batched, dimension checked, retry/backoff; track model version.
Notifications/reminders	Outbox event, preference check, provider delivery attempt, retry and failure state.
SLA scheduler	Periodic calculation using business calendar and pause states; idempotent alerts; avoid duplicate warnings.
Excel exports	Async for large result sets; permission recheck; temporary object storage and expiry.
AI agent execution	Persist workflow state; bounded step count; approval token; idempotent tool actions; recoverable partial failure.

Backup, recovery and portability

- Automated PostgreSQL backups and point-in-time recovery where supported; encrypt backups; test restore on a schedule.
- Back up object storage and verify object/database consistency; define RPO/RTO with product owner.
- Document tenant-level export of core business records and attachments under permissions.
- Provide health/readiness endpoints; graceful shutdown; queue drain; migration rollback/forward strategy.

23. Testing strategy and acceptance

Test level	What to test	Required tooling / practice
Domain unit tests	Invariants, status transitions, money/currency rules, conversion and approval policies.	xUnit; deterministic clock; no DB.
Application tests	Use-case behavior, validation, permissions, tenant context, event publication.	xUnit + mocks/fakes; test negative authorization paths.
Infrastructure integration	EF mappings, constraints, indexes, RLS, transactions, outbox, pgvector queries.	Testcontainers PostgreSQL (+ pgvector image); RabbitMQ/Redis where needed.
API integration	Authentication, tenant resolution, authorization, ProblemDetails, pagination and endpoints.	WebApplicationFactory + Testcontainers.
Angular tests	Components, forms, guards, API error mapping, table filters and permission-driven UI.	Angular test runner; mock HTTP.
End-to-end	Lead-to-opportunity-to-proposal; ticket lifecycle; portal isolation; RAG and approval flows.	Playwright with seeded tenants/users.
Architecture tests	Dependency direction, module boundaries, no Infrastructure references in Domain.	NetArchTest or equivalent.
AI evaluations	Groundedness, citation, retrieval relevance, abstention, tool safety, cost/latency.	Versioned golden dataset + repeatable evaluation runner.

Mandatory security and UAT scenarios

- Tenant A user cannot access Tenant B records via UI, direct API ID, search, reports, export, background jobs, vector retrieval or AI tools.
- Sales executive cannot access another team's records unless role/scope allows it; support agent cannot read unrelated commercial pipeline by default.
- Portal user cannot view internal notes, private attachments, another account's tickets or unpublished knowledge.
- Draft, archived, superseded and unauthorized knowledge never appears in RAG; revoked permissions take effect at query time.
- RAG answer includes valid source title/page/section; unsupported question yields explicit insufficient-information answer.
- AI cannot execute arbitrary SQL, invent tool names, bypass permissions or perform consequential action without approval.
- Approval payload is immutable/single-use/expiring; record changes between preview and execution invalidate or revalidate action.
- Provider outage/rate limit leaves core CRM create/edit workflows available and shows clear AI-unavailable feedback.
- Proposal sent version remains immutable; Won requires outcome date/final value; Lost requires reason.
- Ticket public reply is visible to customer; internal note is never visible externally; SLA pause/resume is correct.

Business acceptance reference

Use BRD UAT-001 through UAT-016 as minimum business acceptance scenarios: lead creation/assignment; duplicate review; lead conversion; opportunity stage tracking; proposal revision/approval; activity recording; ticket resolution; portal isolation; RAG approved-source answer; RAG abstention; AI ticket draft; structured CRM Q&A; agent task preview/approval; provider outage; tenant isolation; and audit trace.

24. Deployment, environments and public demo

Environment	Purpose	Controls
Local	Developer workstation with Docker Compose.	PostgreSQL+pgvector, Redis, RabbitMQ, API, Worker, Angular; seeded synthetic tenant.
Development	Shared integration environment.	CI deployment, test credentials, synthetic data, debug logs restricted.
Staging/UAT	Release candidate and acceptance testing.	Production-like topology; sanitized/synthetic data; feature flags; migration rehearsal.
Production/demo	Public portfolio demonstration.	Synthetic dataset only; strict rate limits; reset mechanism; no real customer data; monitored spend.

Deployment components

- Angular static assets served via CDN/static hosting or reverse proxy.
- ASP.NET Core API behind HTTPS reverse proxy/load balancer.
- .NET Worker Service as separate process/container.
- Managed PostgreSQL with pgvector; Redis; RabbitMQ (managed or operated); object storage; secret manager.
- Central logs, metrics, traces, alerts and dashboards.
- CI pipeline: restore/build → format/analyzers → unit tests → architecture tests → integration tests → frontend tests → security/dependency scan → publish artifact → deploy staging → smoke tests → approved production release.

Public demo requirements

- Demo account with clear role and synthetic IT services company data.
- Visitor can inspect sample data, create/update a lead, move an opportunity, view customer history, create a ticket, search knowledge and try AI assistant.
- Demo data reset mechanism; per-user/session limits; no private customer information; no unrestricted uploads or arbitrary AI actions.
- Display generated content labels, citations and approval previews. Provide a demo walkthrough and architecture/AI evaluation overview.

25. Micro-phase implementation roadmap

Phases are intentionally small. A phase is complete only when its exit gate is met. Sequence can be adjusted only through an explicit dependency decision. Avoid implementing all CRUD screens in one giant phase.

ID	Micro-phase	Implementation work	Exit gate
P000	Repository and delivery baseline	Create Git repo, branch strategy, README, coding standards, editor config, .gitignore, license inventory and issue/phase board.	Repo builds or contains documented empty scaffold; no secrets committed.
P001	Architecture decision records	Record modular monolith, Clean Architecture, tenancy, auth, API versioning, database, AI provider abstraction and deployment decisions.	ADRs reviewed and versioned.
P002	Local developer environment	Docker Compose for PostgreSQL+pgvector, Redis, RabbitMQ; health checks and named volumes.	One command starts dependencies; health checks pass.
P003	Create .NET solution	Create API, Domain, Application, Infrastructure, Worker and Contracts projects.	Solution builds; dependency graph is correct.
P004	Create test projects	Add Domain, Application, Infrastructure, API and Architecture tests.	All test projects run in CI.
P005	Create Angular workspace	Create Angular app, PrimeNG, Tailwind, lint/test config and base routes.	App builds and displays shell.
P006	Shared backend conventions	ProblemDetails, correlation ID, API versioning, Swagger, validation pipeline and exception middleware.	Standard success/error responses tested.
P007	Shared frontend shell	Authenticated layout, navigation, breadcrumb, toast, loading and error components.	Responsive shell works.
P008	EF Core baseline	DbContext, naming conventions, UTC handling, configurations, migration assembly and initial migration.	Migration applies to empty local DB.
P009	Tenant entity and seed	Tenant aggregate, tenant settings, seed/demo tenant and tenant context abstraction.	Tenant created and loaded from trusted context.
P010	Tenant resolution middleware	Resolve tenant from auth claim/host mapping; reject untrusted tenant override.	API tenant context tests pass.
P011	Tenant query filter	Apply tenant filters to tenant-owned entities and safe bypass policy.	Cross-tenant integration tests pass.
P012	PostgreSQL RLS defense	Implement transaction-scoped tenant context and RLS policies where selected.	RLS tests prove isolation under raw SQL too.
P013	Identity integration	ASP.NET Core Identity tables, user entity, password policies and login.	Login/invalid login tests pass.
P014	JWT and refresh sessions	Access token, rotating refresh token, revocation, logout and refresh.	Token lifecycle tests pass.
P015	Current user API	GET /me, tenant, roles, permission claims and profile.	Frontend can load current user safely.
P016	Permission model	Permission keys, role grants, authorization handlers and seed roles.	Permission matrix tests pass.
P017	User administration API	Invite/create/update/deactivate users and list/detail endpoints.	Admin-only API tests pass.
P018	User administration UI	Users grid, filters, invite/edit forms, deactivate and Excel export.	End-to-end invite/manage flow works.
P019	Role management API/UI	Role templates, custom roles and permission matrix.	Least-privilege and privilege escalation tests pass.
P020	Teams management	Teams, memberships, managers and team filters.	Team scoping tested.
P021	Tenant settings	Profile, timezone, currency, locale, branding and business defaults.	Settings persisted and validated.
P022	Audit foundation	Audit event model, write pipeline, query endpoint and sensitive action coverage.	Audit events verified for user/role changes.
P023	File storage abstraction	Object storage adapter, metadata, signed upload/download and scan state.	Cross-tenant file access blocked.
P024	Shared Angular API layer	Typed API clients, auth interceptor, tenant context handling, error mapping and route guards.	Auth/403/409/network scenarios handled.
P025	Reference data framework	Tenant-configurable categories/statuses/sources/priority values.	Reference values can be managed and deactivated.
P026	Lead domain model	Lead entity, source/status/rating, validation and EF config.	Domain tests for lead invariants pass.
P027	Lead create API	Create lead command, validation, duplicate warning query and audit.	Create API integration tests pass.
P028	Lead list API	Server paging/sorting/filtering and owner scope.	Filter/sort/tenant tests pass.
P029	Lead list UI	Grid, search, filters, Add New, View/Edit and Excel export.	Lead list E2E works.
P030	Lead create/edit UI	Reactive form, validation, owner/team, tags and notes.	Form validation and save flow works.
P031	Lead detail and timeline	Detail tabs, activities and audit links.	Timeline is permission-filtered.
P032	Lead assignment	Assignment command, history, notifications and manager scope.	Assignment audit and permission tests pass.

ID	Micro-phase	Implementation work	Exit gate
P033	Duplicate detection	Exact email/domain and configurable fuzzy candidate generation; compare screen.	No automatic merge; candidates explain match signals.
P034	Lead conversion	Atomic account/contact/opportunity mapping and history preservation.	UAT lead conversion passes.
P035	Account domain and schema	Account aggregate, relationship status, indexes and configs.	Domain tests pass.
P036	Account list/detail APIs	Search/filter/paging, Customer 360 related record queries.	Permission and query tests pass.
P037	Account list/detail UI	Grid, filters, create/edit, detail tabs and timeline.	Account workflow E2E passes.
P038	Contact domain/API	Contact, roles, account association and duplicate checks.	Contact rules tested.
P039	Contact UI	Grid, forms, detail, primary contact management.	Contact workflow E2E passes.
P040	Opportunity/pipeline schema	Opportunity, pipeline, stages, transitions, history and loss reasons.	Transition invariants tested.
P041	Opportunity API	CRUD, list filters, stage change, won/lost commands.	Won/Lost validation and audit pass.
P042	Opportunity list UI	Grid, filters, forms, detail and Excel export.	Opportunity list E2E passes.
P043	Pipeline board UI	Stage columns, cards, accessible list alternative and stage change dialog.	Server validates drag/drop transition.
P044	Opportunity stale detection	Configurable inactivity rule and scheduled evaluation.	Stale indicator is deterministic.
P045	Activities domain/API	Task, interaction, participant, reminder, linked record.	Link permission and due-date tests pass.
P046	Activities UI	My Work, due today, upcoming, overdue, calendar/list and create/edit.	Task completion and filters E2E pass.
P047	Reminder worker	Scheduled reminders, preference checks, outbox and retry.	No duplicate reminder on retry.
P048	Proposal domain/schema	Proposal, version, sections, approval and send records.	Immutable sent version tests pass.
P049	Proposal APIs	Create/version/review/approve/reject/send-response commands.	Approval policy and permissions pass.
P050	Proposal UI	List, editor, preview, version compare and approval screens.	Proposal UAT passes.
P051	Dashboard KPI definitions	Document formulas, date basis, cohort and currency behavior.	Metric definitions approved.
P052	Dashboard APIs	Role-filtered KPI and chart queries.	Reconcile against source records.
P053	Dashboard UI	Cards, charts, date filters and drill-through.	Role-based dashboard E2E passes.
P054	Ticket domain/schema	Ticket, message visibility, assignment, queue, category, priority, status and resolution.	Public/internal separation tests pass.
P055	Ticket APIs	Create/list/detail/reply/internal note/assign/escalate/resolve/reopen.	Permission and tenant tests pass.
P056	Ticket queue UI	Queue grid, filters, views, actions and Excel export.	Support workflow works.
P057	Ticket workspace UI	Conversation, internal notes, attachments, SLA header and related tickets.	Portal cannot see internal data.
P058	SLA configuration	Business calendar, holidays, policy, pause states and target calculation.	Boundary/timezone tests pass.
P059	SLA scheduler and alerts	At-risk/breach evaluation, notifications and escalation events.	Idempotent alerts and UAT verified.
P060	Customer portal identity	Separate portal route/app, invitation and account/contact binding.	Portal user scope verified.
P061	Portal ticket pages	Home, ticket list/detail, public reply and attachments.	UAT portal isolation passes.
P062	Knowledge article domain	Article lifecycle, versions, classification, owner and access rules.	Draft/published/archive rules tested.
P063	Knowledge article UI/API	Article list, editor, review/publish, versions and Excel metadata export.	Publishing workflow E2E passes.
P064	Document ingestion registration	Upload validation, metadata, job table and storage.	Unsafe file rejected.
P065	Text extraction	PDF/DOCX/TXT extraction, page/heading metadata and normalization.	Extraction fixtures pass.
P066	Chunking service	Heading-aware token-aware chunker, overlap, metadata and version.	Chunk quality fixtures reviewed.
P067	Embedding provider adapter	IEmbeddingGenerator abstraction, model config, batch/retry and dimensions.	Provider-independent tests pass.
P068	pgvector persistence/search	Vector column, index, keyword search and hybrid candidate query.	Tenant/ACL query tests pass.
P069	Knowledge ingestion worker	Queue processing, retries, status, atomic activation and reindex.	Failed job recoverable.
P070	Knowledge search UI	Keyword/hybrid search, filters, result detail and source location.	Only authorized published docs returned.
P071	RAG retrieval service	Query embedding, hybrid retrieval, ACL recheck, reranking hook and citations.	Golden retrieval set passes.
P072	LLM gateway	Chat model adapter, provider config, timeout/retry/rate-limit and fallback.	Core CRM unaffected by provider outage.
P073	Prompt registry	Versioned prompts, variables, release metadata and test fixtures.	Prompt version captured per run.

ID	Micro-phase	Implementation work	Exit gate
P074	AI conversation persistence	Conversation/message/run models, ownership and retention.	Cross-user conversation access blocked.
P075	AI chat API and streaming	Message endpoint, streaming response and structured response events.	Cancel/retry and error handling work.
P076	AI assistant UI	Chat workspace, citations, feedback, streaming, record links and history.	AI chat E2E passes.
P077	RAG answer generation	Grounded response prompt, citation validation and abstention.	UAT RAG answer/abstention passes.
P078	AI safety controls	Prompt injection tests, content boundaries, redaction, quotas and tool isolation.	Safety regression suite passes.
P079	Structured CRM tools	Search leads/opportunities/tickets, account summary and KPI query tools.	Tool output matches API query and permissions.
P080	Structured CRM Q&A;	Intent routing, validated arguments, deterministic tool calls and response rendering.	No arbitrary SQL; query parity tests pass.
P081	Ticket AI classification	Category/priority suggestion, confidence and manual override.	Low-confidence/manual fallback tested.
P082	Ticket AI summarization	Conversation summary, attempted steps and outstanding questions.	Factual consistency review passes.
P083	Ticket RAG assistance	Similar tickets, KB retrieval and cited draft reply.	No unsupported fix; no auto-send.
P084	Lead/account/opportunity summaries	Contextual summary and evidence links.	No unauthorized records in summary.
P085	Meeting note assistant	Extract summary, decisions and action items from provided notes.	Human review before saving.
P086	Agent workflow framework	Bounded state machine, step/time/token budgets, run persistence.	Timeout/retry/recovery tests pass.
P087	Agent approval UX/API	Preview, edit, approve/reject, expiry and payload hash.	Only authorized approval executes.
P088	Stalled opportunity agent	Identify stale deals, evidence and proposed next action/task.	No task created before approval.
P089	Customer 360 agent	Assemble permitted account facts and cite record links.	Read-only; date coverage shown.
P090	Ticket analysis agent	Summarize issue, steps tried, unanswered questions and sources.	No automatic diagnosis or reply.
P091	Agent tool execution	Idempotency, revalidation, audit, partial failure and retry.	Duplicate actions prevented.
P092	AI usage/cost dashboard	Tokens, spend estimate, model, latency, errors and quotas.	Tenant admin view permissioned.
P093	AI evaluation harness	Golden RAG/structured/tool datasets and regression runner.	Release thresholds recorded.
P094	AI feedback and quality reports	Helpful/unhelpful, accepted/edited/rejected and groundedness review.	Feedback not mistaken for correctness.
P095	Standard reports	Lead funnel, pipeline, win/loss, activity, account and support reports.	Metric reconciliation passes.
P096	Excel export service	XLSX generation, filters, async job, audit, expiry and formula-injection defense.	Tenant and row permission rechecked.
P097	Notification center	In-app notifications, preferences, read/unread and deep links.	No duplicate delivery.
P098	Email adapter	Invite/reminder/approved outbound email; template and delivery status.	No AI auto-send.
P099	Audit viewer	Search/filter audit events and export if permissioned.	Sensitive data not leaked.
P100	Performance and query tuning	Indexes, pagination, query plans, load tests and caching.	Targets agreed and met.
P101	Accessibility and UX polish	Keyboard, focus, contrast, responsive layout and empty/error states.	Accessibility checklist passes.
P102	Backup/restore rehearsal	DB/object backup, restore, RPO/RTO documentation.	Restore test evidence stored.
P103	CI/CD pipeline	Build/test/security scan/container build/staging deploy/smoke test.	Protected main branch gates pass.
P104	Staging UAT and security review	Run UAT-001..016, tenant tests, AI red team and permission matrix.	No critical/high findings open.
P105	Synthetic demo dataset	Coherent IT services tenant, users, accounts, leads, deals, tickets and KB.	No real personal/customer data.
P106	Demo reset and rate limits	Safe reset job, quotas, abuse controls and monitoring.	Demo cannot be damaged permanently.
P107	Documentation and handover	Architecture, runbook, API docs, user guide, AI evaluation report and walkthrough.	Another developer can run and understand product.

26. Vibe-coding workflow and prompt contract

Required loop for every micro-phase

- 1. Select exactly one phase and read its prerequisites and exit gate.
- 2. Ask the coding agent to inspect the repository and report the files, patterns and dependencies it plans to touch. Do not allow edits yet.
- 3. Provide the phase requirement, relevant module specification and architecture rules. Ask for a short implementation plan and risks.
- 4. Implement only that phase. Ask the agent to avoid unrelated refactoring and to list every changed file.
- 5. Run formatter, analyzers, build, unit tests, integration tests applicable to the phase and frontend tests.
- 6. Review generated code: tenant scope, permissions, validation, transactions, audit, logging, secrets, query efficiency and error behavior.
- 7. Manually test the UI/API workflow and negative permission cases.
- 8. Update phase checklist, API/schema docs and ADRs if decisions changed. Commit with a small descriptive message.
- 9. Only then move to the next phase.

Reusable prompt template for a coding agent

```
You are working on CRMPro AI. Follow docs/DEVELOPER_SPECIFICATION.md and the approved ADRs.
Current phase: [PHASE ID + TITLE].
Prerequisites completed: [LIST].
Goal: implement only the scope of this phase.
Constraints:
- Preserve Clean Architecture and module boundaries.
- Domain/Application must not depend on Infrastructure or UI frameworks.
- Enforce TenantId, record-level authorization and audit where applicable.
- Use EF Core Code First; add a reviewed migration for schema changes.
- Use DTOs, validation, ProblemDetails and server-side paging/filtering.
- Add positive, negative, tenant-isolation and permission tests.
- Follow existing naming, formatting and error conventions.
- Do not invent business rules; list open decisions instead.
Before coding: inspect the repository and return a concise plan, affected files and risks.
After coding: list files changed, migration, tests executed/results, manual test steps, known gaps and next
dependencies. Do not claim tests passed unless executed.
```

Definition of Done

- Scope matches the phase and BRD; no hidden scope expansion.
- Build, formatting, analyzers and required tests pass.
- Tenant and authorization tests include forbidden and cross-tenant cases.
- Database migration reviewed; no production auto-migration.
- UI has loading, empty, error, validation and success states.
- Audit, logging, correlation and privacy requirements implemented.
- API/OpenAPI and developer docs updated.
- Manual verification steps and evidence recorded; code committed.

27. Open decisions and assumptions register

The BRD explicitly identifies several decisions for product-owner confirmation. These must not be silently guessed by a coding agent.

Decision	Proposed default for build	Owner / status
Target customer size and plans	Start with 5–200 internal users as planning assumption; no billing/subscription engine in MVP.	Product owner – confirm.
Lead qualification fields and status flow	Configurable statuses; use source, need, budget, authority, timing, owner and next action.	Sales manager – confirm.
Opportunity pipeline stages/transitions	Tenant-configurable pipeline with ordered stages and transition rules.	Sales manager – confirm.
Currency handling	Store currency per opportunity/proposal; aggregate by currency unless an approved FX source/rule exists.	Business owner – confirm.
Proposal approval policy	Configurable by amount/discount/role; default manual review before send if required.	Sales manager – confirm.
Ticket statuses, priorities and SLA	Tenant-configurable; define business hours, holidays, pause states and resolution codes.	Support manager – confirm.
Portal visibility	Portal user sees own organization tickets and only public replies/shared attachments.	Product owner – confirm.
AI providers and regions	Provider selected after privacy, region, price, latency, data retention and capability review.	Organization owner – confirm.
AI retention and training	Document provider terms; minimize context; define CRM conversation/log retention.	Privacy reviewer – confirm.
AI action approval policy	Require explicit approval for external messages and material record/customer commitment changes.	Business owner – confirm.
Document size/types and retention	PDF/DOCX/TXT; configure size cap, malware scanning and retention.	Admin – confirm.
Email/calendar integrations	Not assumed for MVP beyond transactional email; calendar sync is optional future scope.	Product owner – decide.
Predictive ML readiness	Keep lead/deal scoring disabled until adequate labeled historical outcomes and evaluation exist.	Product owner/data owner – later.
RPO/RTO and backup retention	Set based on hosting budget and business criticality; rehearse restore.	Technical owner – confirm.
Deletion/merge policy	Archive by default; merge requires compare, confirmation and history preservation.	Product owner – confirm.

Assumptions used in the detailed UI specification

- List screens use server-side pagination, filtering and sorting. Exact default page size is configurable (suggest 25).
- All module lists include Add New, View/Edit and permission-controlled archive/delete where applicable; destructive operations are not universally exposed.
- Excel export is the only user-facing export format. Export is permission-controlled, audited and filtered server-side.
- Business statuses and categories are configurable tenant data unless the BRD requires fixed technical states.
- Grid columns and form fields in this document are proposed baseline fields derived from the BRD's business concepts; confirm exact requiredness and terminology with stakeholders.

28. BRD traceability and final release checklist

BRD area	Developer specification coverage
Organization setup / tenancy	Sections 3, 4, 19, 22 and phases P009–P025.
Users, roles and access	Sections 3, 4, 22 and phases P013–P024.
Dashboard and reporting	Sections 5, 14, 20 and phases P051–P053, P095–P096.
Lead management	Section 6 and phases P026–P034, P081, P084.
Accounts and contacts	Section 7 and phases P035–P039, P084, P089.
Opportunity pipeline	Section 8 and phases P040–P044, P088.
Activities and communications	Section 9 and phases P045–P047, P085.
Proposals	Section 10 and phases P048–P050.
Tickets and SLA	Section 11 and phases P054–P059, P081–P083, P090.
Customer portal	Section 12 and phases P060–P061.
Knowledge and documents	Section 13 and phases P062–P071.
AI assistant, RAG and agents	Sections 15–18 and phases P067–P094.
Audit, operations and demo	Sections 3, 22–24 and phases P022, P097–P107.

Release gate checklist

- All Must requirements are implemented or explicitly deferred with product-owner approval.
- Core lead-to-customer lifecycle and ticket support journey complete without manual database intervention.
- Every active lead, opportunity and ticket has visible ownership and status.
- Tenant isolation and role/record permissions pass automated and UAT scenarios.
- AI outputs are labeled generated/suggested, cite evidence where applicable, abstain when unsupported and have manual fallbacks.
- Support can operate without AI; provider outage does not block core CRM.
- Reports reconcile to approved metric definitions.
- Exports are XLSX only, audited and permission-filtered.
- Public demo uses synthetic data, safe rate limits and reset capability.
- Architecture, user guide, runbook, API docs, AI evaluation report and demo walkthrough are complete.

Source and implementation note

Business baseline: CRMPro AI Business Requirements Document, Version 1.0 (34 pages; prepared 06 November 2026). This developer specification preserves its product boundary, stakeholder model, functional module intent, Must/Should/Could prioritization, AI safety principles, non-functional requirements, UAT scenarios and staged release recommendation. Technical details not explicitly fixed by the BRD are marked as proposed implementation baselines or open decisions.

Do not treat this document as a substitute for stakeholder sign-off. Before production, approve the open decisions, security/privacy posture, AI provider terms, data retention, SLA definitions and operational RPO/RTO.